

Evaluation of Three In-Vehicle Interactions from Drivers' Driving Performance and Eye Movement Behavior *

Guofa Li, Fangping Zhu, Tingru Zhang, Ying Wang, Shengfan He, Xingda Qu

Abstract— Although in-vehicle interaction technologies offer assistances for drivers from multi-aspects, the distractions that may cause by these technologies would significantly challenge driving safety. This study examined the influences of three in-vehicle interactions on drivers' driving performance and eye movement behavior. The three in-vehicle interactions include phone pick-up (PPU), Bluetooth and audio (BT+AUD) and Bluetooth and head-up display (BT+HUD). A control group (i.e., None, in which no human-machine interaction was included while driving) was designed to highlight the influences of the examined interactions. Fourteen licensed drivers participated in the simulation experiments. Their age ranged from 20 to 30 years old (Mean $\pm$ SD = 25.1 $\pm$ 3.8). They were required to complete the Location Based Service (LBS) tasks using the three interactions while driving. Both driving performance and eye movement data were collected for analysis. Results showed that, when using PPU or BT+HUD, the minimum throttle opening, the steering wheel angle (SWA) and the Shannon entropy (SE) of SWA were all greater than those values when using BT+AUD or None. As for the SE of throttle opening and all the examined eye movement features (frequency and duration of fixation/saccade), the values when using BT+AUD or BT+HUD were higher than the values when using PPU or None. The presented results in this study provide evidences for car manufactures to design better BT+AUD human-machine interaction systems as a priority strategy.

Index Terms— Driving performance, eye movement, in-vehicle interactions, human factors, driving behavior

I. INTRODUCTION

With the rapid development of in-vehicle interactive technologies and the accessibility to portable electrical devices, there is an increasing trend for drivers to engage in non-driving-related tasks (NDRTs) while driving [1-4]. NDRTs can lead to driver distraction or inattention by competing for a variety of resources (e.g. visual, manual or cognitive) with the major driving task. According to [5], distraction associated with NDRTs has become one major cause of road traffic accidents. Therefore, minimizing the distracting effect of NDRTs is of great importance to today's road safety. This can be achieved by improving the interface used when performing NDRTs [6].

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The most common interaction when performing NDRTs is the visual-manual interaction. Yet such interaction is not always safe considering that it usually requires drivers to take their eyes off the road to visually read messages, and to take their hands off the steering wheel to operate buttons. It has been well documented that, when engaging in visual-manual based NDRTs, drivers tend to have longer reaction times [7], reduced longitudinal and lateral control [8], and a higher scanning randomness [9]. These adverse effects on driving promotes the need for less distracting interactions, thus speech-based system have been introduced as a possible solution.

The speech-based interaction has become available with the development of voice-recognition technologies. It has an advantage over visual-manual interaction by minimizing the manual manipulation of input devices and glances away from the road. Speech-based interaction has been consistently reported as less disruptive to driving than visual-manual based interaction in the driving literature [10][11]. For instance, He et al. found that drivers showed better vehicle controlling performance when entering text messages through speaking than through typing [11]. Using meta-analysis, Simmons et al. reviewed studies that had compared the effects of two interaction and concluded that drivers had better performance with the speech-based system on reaction times, lane positioning and headway [12].

It should be noted that [12] have assumed that the speech-based systems from different studies all had similar effect size although these systems are not exactly the same. However, recent evidence suggested that different types of speech-based systems varied in their influences on driving performance. Reimer et al. carried out a simulated driving experiment to investigate whether accessing speech-based interaction through portable devices and embedded systems had similar distracting effects [13]. In either way, drivers needed to press a button to activate the system first and then vocally entered a dialing command. Their results suggested that embedded vehicle systems advantaged over portable device across a number of metrics such as lower workload, fewer task errors and shorter task completion time. However, their effects on driving performance had not been directly investigated.

While speech-based interface seems to be a promising solution to the distraction caused by NDRTs engagement, it is not totally distraction-free [14-16]. This is especially true when using speech feedback as the output of human-machine interaction. First, understanding the speech output can pose cognitive demand that could undermine driving performance. For instance, listening to emails while driving increased driver reaction time by 30% [14]. Another disadvantage associated with the speech-based output is that it usually takes longer

time for humans to acquire auditory messages than visual message that contains the same information. This raises the question whether a new interface that using speech as input and visual feedback as output can over perform the traditional speech-in and speech-out system. Wang et al. has carried out some preliminary work and found that the speech-in and visual-output interface resulted in a shorter task completion time compared with speech-in and speech-out interface [6]. However, how this novel interface affects driving performance has not been explored in their study.

To sum up, although previous literatures suggested speech-based interactions are promising ways to render NDRTs, their effects on driving performance and visual search behavior have not been understood well. This study aims to partially fill up this gap by investigating how three promising interaction modes would influence vehicle manipulation as well as attention distribution.

The remainder of this paper is organized as follows: Section II introduces the details about the simulation experiment and the analyzed data. Section III presents the results found in this study. Section IV discusses the advantages and disadvantages of each interaction type by explaining why the interaction type leads to the better or worse results. Section V concludes this paper and proposes suggestions on the possible applications and future efforts.

II. METHOD

A. Apparatus

The simulation experiments presented in this study were conducted on a fixed-base driving simulator in Beihang University, Beijing, China. See Figure 1 for the simulator. The simulated roadway design and traffic scenarios were generated by UC-Win/Road Ver.8 Standard (FORUM8, Tokyo, Japan) with a refreshing rate of 20 Hz. The participants were required to complete a Location Based Service (LBS) driving task using different interactions while driving. An Apple iPhone 5s (screen size: 4 inches; resolution: 1136 x 640) operated by iOS 7 was used to conduct the LBS tasks. Smart Eye Pro Revision 5.9 was used to collect drivers' eye movement data.



Figure 1. The used driving simulator in this study

B. Participants

Fourteen licensed drivers including twelve males and two females participated in this study. Their age ranged from 20 to 30 (Mean $\pm$ SD = 25.1 $\pm$ 3.8), which ensured that the participants were familiar with the usage of smart phones. A consent form approved by the local ethics committee was

signed prior to the experiments. Each participant was paid 100 RMB as a compensation for their efforts involved in the experiment.

C. Interactions and Tasks

Three human-machine interaction types were examined in this study, i.e., phone pick-up (PPU), Bluetooth and audio (BT+AUD), and Bluetooth and head-up display (BT+HUD). The Bluetooth module was flexibly mounted on the steering wheel with a preferred position by the driver. As long as an iPhone with the Bluetooth module was connected, *Siri* could be simply activated by pressing the button mounted on the steering wheel (See Figure 2).

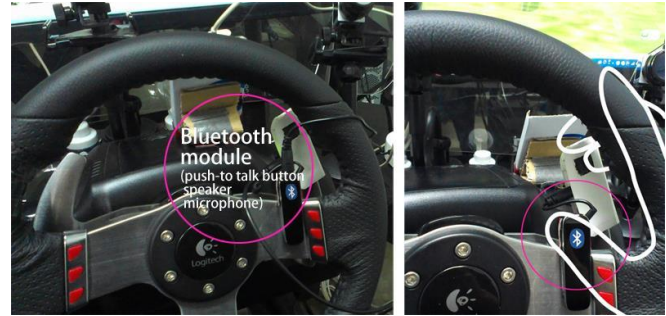


Figure 2. The Bluetooth module mounted on the steering wheel

Using PPU, BT+AUD or BT+HUD, the LBS task required drivers to search a designated place (e.g., gas station, restaurant, hotel, etc.), select a destination with a designated criterion (e.g., the nearest), and present the navigation route. For example, a given LBS task would be 'please find the nearest gas station'. The details on how to use the three interactions to complete the LBS task are listed below:

(1) Phone pick-up (PPU)

Drivers use PPU to complete an LBS task following the steps: 1) Drivers press the button to activate *Siri*; 2) Drivers tell *Siri* what they want to search; 3) Drivers wait for the search results to present on the iPhone screen; 4) Drivers select the target destination on the screen to wait till the navigation route is ready.

(2) Bluetooth and audio (BT+AUD)

Drivers use BT+AUD to complete an LBS task following the steps: 1) Drivers press the button to activate *Siri*; 2) Drivers tell *Siri* what they want to search; 3) *Siri* reads the search results one by one, and drivers check each option by telling *Siri* 'Next' if the current option is NOT the desired destination or 'Go there' if it is the desired one; 4) *Siri* prepares and generates the navigation route.

(3) Bluetooth and head-up display (BT+HUD)

Drivers use BT+HUD to complete an LBS task following the steps: 1) and 2) are the same with the first two steps when using BT+AUD; 3) *Siri* presents the search results list on the head-up display area of the windshield; 4) Drivers tell *Siri* the rank of the target destination on the list (for example, drivers would say 'Go to the Second'), and *Siri* prepares and generates the navigation route. See Figure 3 for an overview of how the BT+HUD interacts with drivers while driving.

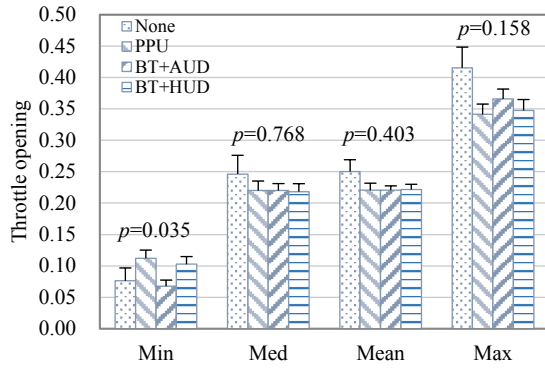


Figure 4. Min, Med, Mean and Max values of throttle opening when using different interactions. (The error bars represent standard errors.)

As for the lateral driving performance, the Min, Med, Mean and Max values of SWA when using PPU or BT+HUD were much higher than the values when using BT+AUD or None. See Figure 5. Statistical significances were found in the Min ($p=0.005$), Med ($p<0.001$) and Mean ($p<0.001$) of SWA. Even no significant effect was found in the Max of SWA, the value when using PPU was 360% higher than the value in the None situation ($p=0.026$). Similarly, the Max of SWA when using BT+HUD was 4.4 times of the number in the None interaction group where no LBS tasks were required ($p=0.034$). See Table III for pairwise comparison results to check the statistical significance between each two of the interaction groups.

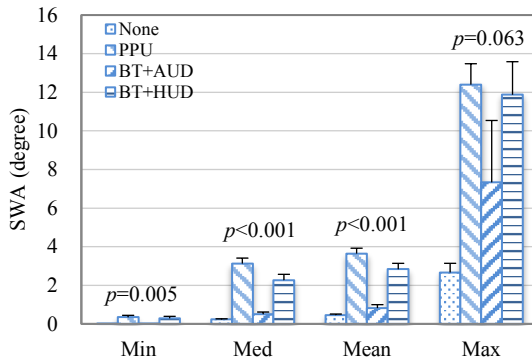


Figure 5. Min, Med, Mean and Max values of SWA when using different interactions.

TABLE III. PAIRWISE COMPARISON RESULTS OF THE SWA FEATURES

		None	PPU	BT+AUD
Min	PPU	0.032	-	-
	BT+AUD	0.985	0.002	-
	BT+HUD	0.07	0.616	0.009
Med	PPU	<0.001	-	-
	BT+AUD	0.577	<0.001	-
	BT+HUD	<0.001	0.01	<0.001
Mean	PPU	<0.001	-	-
	BT+AUD	0.467	<0.001	-
	BT+HUD	<0.001	0.02	<0.001
Max	PPU	0.026	-	-
	BT+AUD	0.282	0.091	-
	BT+HUD	0.034	0.86	0.127

As illustrated in Figure 6(a), the SE of SWA in the None interaction group was the lowest, which indicates a stable SWA operation pattern to keep the subject vehicle driving in the lane. The SE of SWA in the BT+AUD interaction group was better than this feature in PPU or BT+HUD. Statistical significance was found in the SE of SWA among the examined interaction groups ($p=0.006$). This better lateral performance was consistent with the results found in Figure 5. Interestingly, the SE of throttle opening in the BT+AUD group ranked the highest among the examined interactions, which indicates that drivers operate the throttle more frequently in the BT+AUD group than in the other groups. Statistical significance was found among the examined interactions ($p<0.001$). See Figure 5(b). The higher SE of throttle operation but lower SE of SWA when using BT+AUD probably indicate that although drivers worked in a high-workload situation (i.e., the LBS task in this study), the BT+AUD interaction could prevent drivers from distracting too much by the NDRTs to keep safe longitudinal and lateral performances.

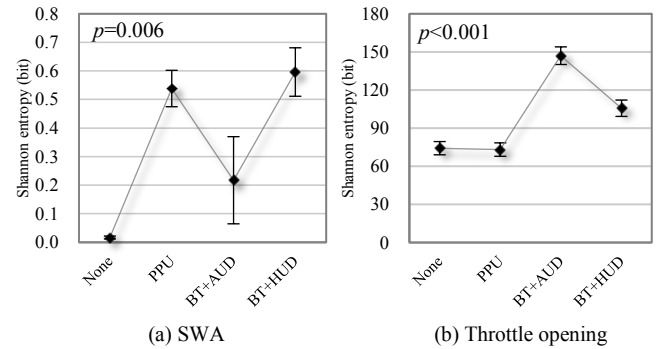


Figure 6. Shannon entropy of SWA and throttle opening

Both the longitudinal and lateral accelerations showed no statistical significances among the examined interaction groups.

B. Eye Movement

To intuitively illustrate the road center area, the fixations fell within this area were marked with a dark red color, while the other fixations were marked with dark blue. See Figure 7. No significant differences were found in the attention allocation in the road center areas when using different interactions. The coordinate of the road center point is presented in Table IV. The RFCPs when using None, PPU, BT+AUD and BT+HUD were 77.6%, 86.5%, 87.4% and 82.9%, respectively. The lowest RFCP in the None interaction group indicates that drivers with a low workload would divide their attention more to the other areas. However, when there was a NDRT that required a relatively higher workload, drivers would concentrate more in the road center area to keep aware of the forward road traffic for safety reasons.

TABLE IV. THE COORDINATE OF ROAD CENTER POINT AND THE RFCP WHEN USING DIFFERENT INTERACTIONS

Interactions	Horizontal coordinate (°)	Vertical coordinate (°)	RFCP
None	-1.16	-2.35	77.6%
PPU	-2.80	-2.69	86.5%
BT+AUD	-0.90	-2.10	87.4%
BT+HUD	-4.29	-1.34	82.9%

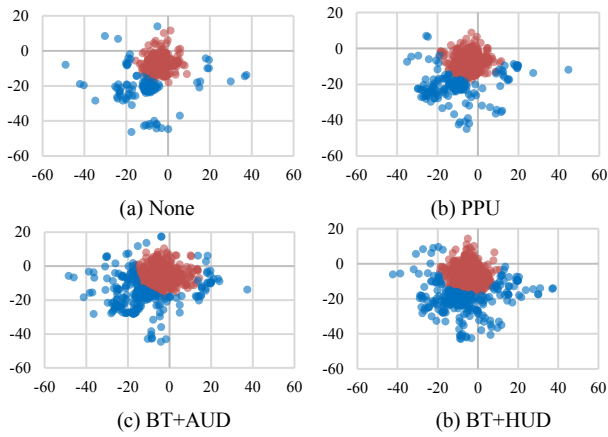


Figure 7. Road center area in the four examined situations (the units of the horizontal and vertical axis are all degree (°))

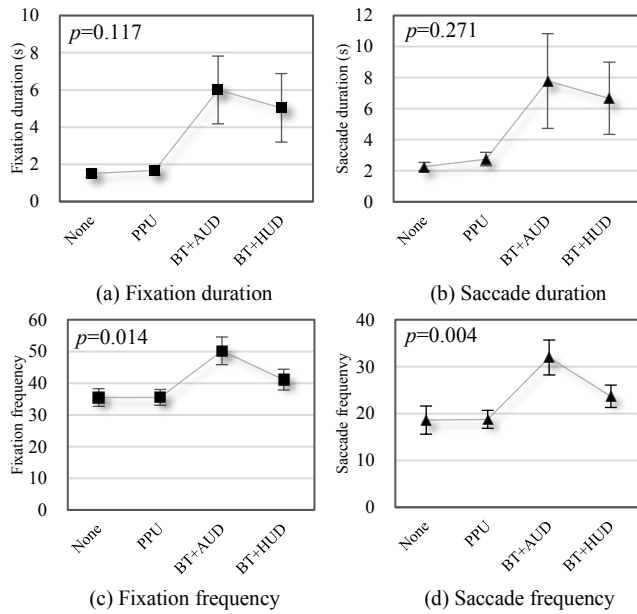


Figure 8. Fixation and saccade features in the four examined interaction groups.

TABLE V. PAIRWISE COMPARISON RESULTS OF THE EYE MOVEMENT FEATURES

		None	PPU	BT+AUD
Fixation duration	PPU	0.954	-	-
	BT+AUD	0.117	0.039	-
	BT+HUD	0.209	0.096	0.637
Saccade duration	PPU	0.906	-	-
	BT+AUD	0.186	0.099	-
	BT+HUD	0.281	0.182	0.708
Fixation frequency	PPU	0.996	-	-
	BT+AUD	0.025	0.002	-
	BT+HUD	0.379	0.224	0.052
Saccade frequency	PPU	0.975	-	-
	BT+AUD	0.012	0.001	-
	BT+HUD	0.326	0.186	0.029

See Figure 8 for the duration and frequency of fixations and saccades when using different interactions. The fixation durations when using BT+AUD and BT+HUD were longer than that in the None and PPU interaction groups. This feature in the None and PPU groups was almost the same, but the value when using BT+AUD was slightly longer than that when using BT+HUD. Similar patterns were found in all the four examined features, i.e. fixation duration/frequency and saccade duration/frequency. Statistical significances were found among the four examined situations in the fixation frequency ($p=0.014$) and saccade frequency ($p=0.004$). See Table V for the pairwise comparison results for the fixation and saccade features.

IV. DISCUSSION

When using PPU, drivers need to move their attention allocation from the surrounding road traffic to the smart phone, which will significantly distract the drivers. The moving distance of drivers' attention allocation is the longest when using PPU. According to the Fitts' Law originally proposed by Paul Maurice Fitts in 1954 [21], when the size of the target area is determined, the time humans needed to transit from the original allocation to the target area is proportional to the transition distance. Besides, drivers also need to move and/or turn their head from the forward road to focus on the smart phone for the LBS task. The head movement would increase the difficulty and workload of LBS task when using PPU compared with the other interactions, thereby the high workload would degrade drivers' situation awareness. Such degraded situation awareness level would easily lead to worse longitudinal and lateral driving performances [22], less frequent fixations and saccades, and shorter durations of the fixations and saccades [18].

When using BT+HUD, neither do drivers need to turn their heads with large angles nor do they need to move their attention allocations with long distances. However, the displayed information on the screen when doing the LBS task will partially occupy drivers' physiological capabilities that are assigned to deal with visual information. The rest capability to deal with road traffic situations will be restricted. Thus, the longitudinal and lateral driving performances will be degraded when using BT+HUD. Whereas the higher frequencies and longer durations of eye movement features indicate that drivers were driving with higher situation awareness levels when doing the LBS tasks using BT+HUD. In conclusion, when doing NDRTs like LBS, BT+HUD can help keep a reasonable situation awareness level but will degrade drivers' driving performance. Similar results were found in [16].

When using BT+AUD, drivers are released from getting NDRT information from the visual channel. Thus, they could concentrate on keeping driving safely to get the best driving performance among the examined interactions. As for the situation awareness levels in the examined interaction groups, both BT+HUD and BT+AUD can help drivers keep aware of the potential dangers. That's because NDRTs would usually make drivers stressed and be more conscious of the surrounding traffic situations, so that they would check with higher frequencies and longer durations. However, the PPU solution overloads drivers so that they cannot keep aware of the traffic situations with appropriate visual search behavior.

The present results in this study prove that BT+AUD could prevent drivers from distracting too much by the NDRTs to keep driving safely. It was also concluded in [23] that adding sound to an in-vehicle interface was helpful on reducing glances toward a visual display, i.e., less distraction.

V. CONCLUSION

This study examined drivers' driving performance and eye movement patterns when searching a required destination on a smart phone in four different interaction situations, i.e., None, PPU, BT+AUD and BT+HUD. Results showed that the Min throttle opening value when using PPU or BT+HUD was greater than that when using BT+AUD or None. Similar results were found for the SWA and the SE of SWA. Concerning the SE of throttle opening and all the examined eye movement features, the values when using BT+AUD or BT+HUD were higher than the values when using PPU or None. To conclude the comparison results concerning the driving performance and eye movement features when using the examined interactions, BT+AUD is the best and BT+HUD ranks the second. PPU is the worst interaction type. The results presented in this study provide evidences for car manufactures to design better BT+AUD human-machine interaction systems instead of BT+HUD systems. A well-designed BT+AUD interaction system could assist drivers without distracting them too much to keep an appropriate workload and situation awareness level while driving.

For future studies, the following topics are suggested to be further considered: (1) This study does not examine the performance of the interaction types on highways. As highway driving is another key element in drivers' practical driving experience, the investigation on the effectiveness of the interactions on highways is a solid requirement [24]. (2) Field operation tests are needed to re-examine the findings concluded from simulation experiments [25], so that a more solid conclusion could be drawn to encourage safe and reliable applications in vehicles.

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